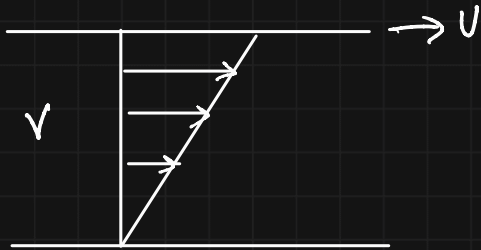




11/1/23

anything associated with movement of mass $\rightarrow$ convective

if two cylinders are very close $\rightarrow$ then they can be opened up and be considered as parallel plates



Couette flow



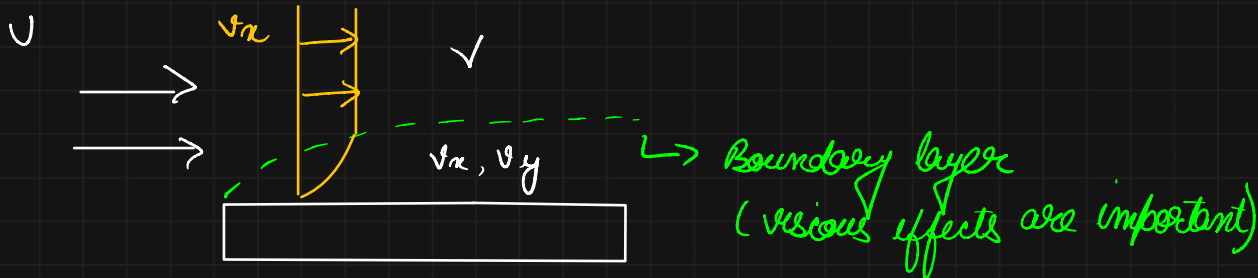
Pressure gradient flow

shear stress $\rightarrow \mu$ times velocity gradient

13/1/23

Order of magnitude analysis of NS equation

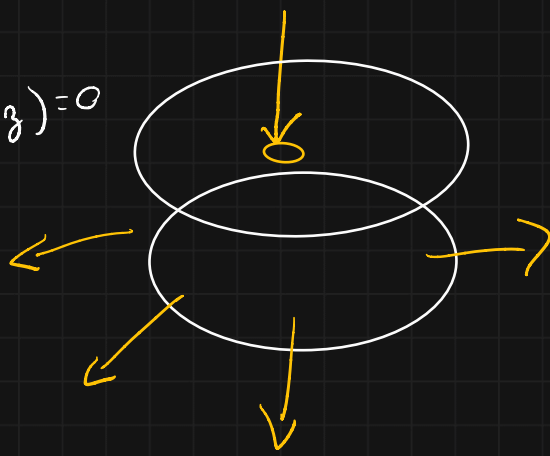
$$\rightarrow \frac{\partial v_x}{\partial x} \approx \frac{v_x}{L_x} \quad \frac{\partial v_x}{\partial y} \approx \frac{v_x}{L_y}$$



governing equation

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right)$$

$$\frac{\partial p}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (r v_\theta) + \frac{\partial}{\partial z} (r v_z) = 0$$



creeping flow $\rightarrow$ role of convection is not important
governing equation

$$\cancel{r} \cancel{v_r} \frac{\partial v_r}{\partial r} = -\frac{dp}{dr} + \mu \frac{\partial^2 v}{\partial z^2}$$

As we are $\leftarrow \approx 0$

just considering creeping flow.

$$\mu \frac{d^2 v}{dz^2} = + \frac{dp}{dr}$$

we have written $\frac{\partial p}{\partial r} \rightarrow \frac{dp}{dr}$

as z is very small, so p won't vary with z

$$P \rightarrow f(r, z)$$

$$P \rightarrow f(r) \rightarrow \text{only due to small } z$$

P is a fn. of z & r but very weak fn. of z

Flow in a pipe

$p_1 \rightarrow$

p_2 flow rate will depend on

$$\left(\frac{p_1 - p_2}{L} \right) \rightarrow \text{is a constant}$$

If not constant then $\rightarrow$ at different points diff flow rates
which will contradict eqn. of continuity

$\rightarrow$ incompressible flow
 $\rightarrow$ cross section area constant $\rightarrow$ reason for $\frac{p_1 - p_2}{L}$ to be constant
 $\hookrightarrow$ linear

25/11

Boundary layer

Boundary layer $\rightarrow$ very thin $\rightarrow$ Important

$\rightarrow$ Inside it viscous forces are predominant

$\rightarrow$ Beyond it inviscid

ex - Racing car moving at high speed $\rightarrow$ "wakes" (low pressure regions) will form and this will lead to more drag, so spoilers are added to break the "wakes" into smaller regions.

its viscosity $\rightarrow$ which creates velocity gradient

$\hookrightarrow$ makes flow 2D [assuming no v_z (in depth)]

The thickness of layer upto which viscous forces effects are dominant



boundary layer

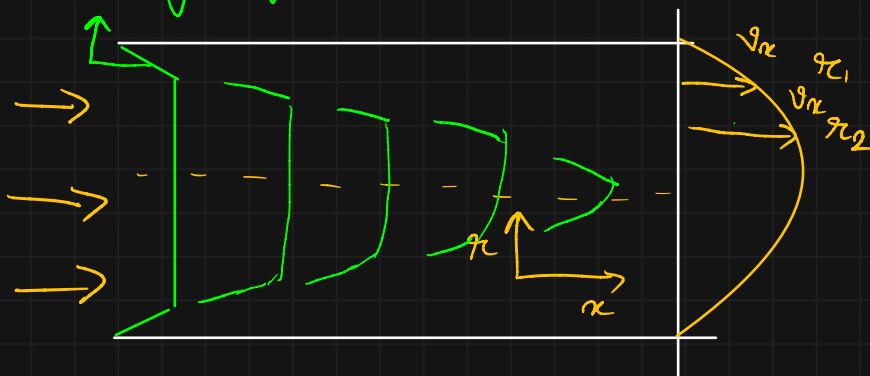
Thickness of BL = 99% of freestream velocity

In order for BL to grow $\rightarrow$ there has to be flow across the BL

$\hookrightarrow v_y \neq 0$

$\hookrightarrow$ which also justifies $v_x = f(x, y)$

velocity profiles



$$\frac{dv_x}{dy} = 0$$

$\downarrow$
fully developed flow

In channel flow $\rightarrow$ when BL merge we get fully developed flow and after that v_x will not vary with y

Flow inside the boundary layer

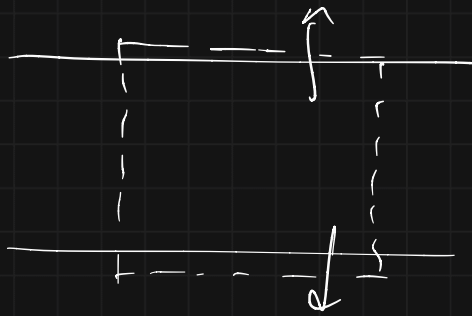
$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right) \rightarrow \text{For laminar flow only}$$

↗ can be neglected

assumptions

$$v_x \gg v_y$$

$$\frac{\partial v_x}{\partial y} \gg \frac{\partial v_x}{\partial x} \rightarrow \text{as } y \text{ is small}$$



In the boundary layer,

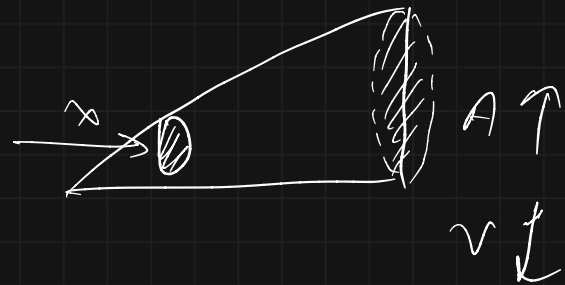
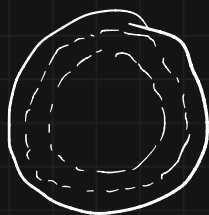
$$v_x = f(\underline{x}, \underline{y})$$

v_x at $x=0$
 $v_x \rightarrow$ profile?

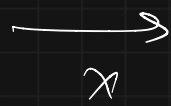
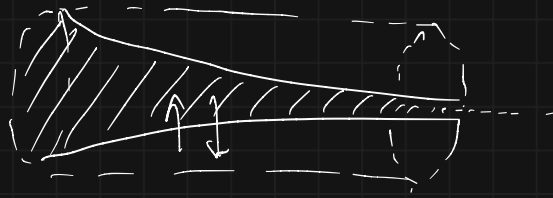


$$v_x < U$$

m.a. +
 f.v



$$\underline{\underline{A \cdot v = R}}$$



Area is changing
 $A \uparrow$

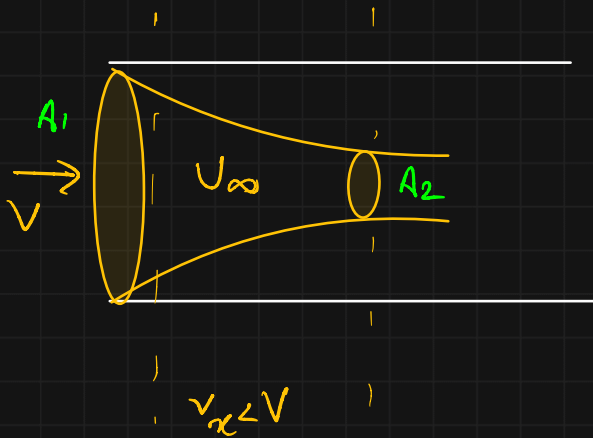
$$\Rightarrow v \downarrow s.$$

3/2/23

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

Boundary conditions

$$\begin{array}{ll} \text{At } y=0 & v_x, v_y = 0 \\ y=\delta & v_x = U_\infty \\ x=0 & v_x = V \end{array}$$



→ As area ↓ ($A_2 < A_1$) → ($U_\infty > V$)

$U_\infty > V$ and will keep on increasing till boundary layers merge

8th July

Momentum integral equation is valid for both turbulent & laminar
just that u will have to decide on ν/μ & τ_w

anything you calculate out of integral equation $\rightarrow$ gives value
"ON THE CV"

For the case of suction $\rightarrow$ At $x=0, y=0$
 $\downarrow$
On the starting otherwise $x=0, y=0 \rightarrow v_x=0, v_y=0$
 $\rightarrow v_x=0$ & $v_y \neq 0$

Reynold's stresses $\rightarrow$ due to mixing, not only viscous

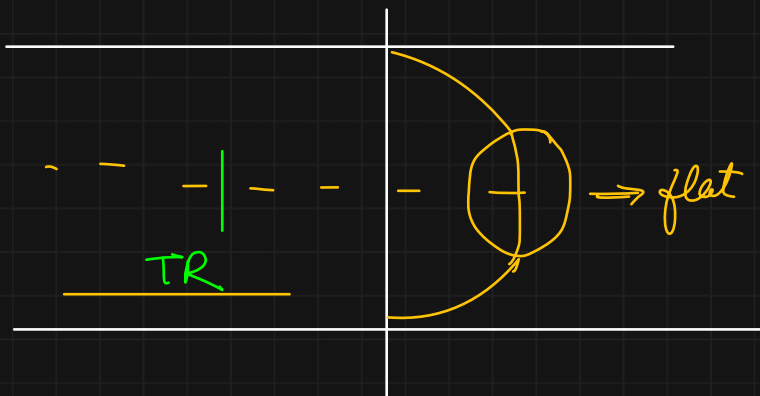
For high turbulent flow

$\rightarrow$ near wall $\rightarrow$ viscous will dominate
 $\rightarrow$ at core $\rightarrow$ " " not dominate

higher degree of turbulence $\rightarrow$ more universal velocity profile

$\downarrow$
In laminar there was a certain point.

$v^* \rightarrow$ friction velocity



Transient conduction $\rightarrow$ spacewise isothermal $\rightarrow$ but varies with time

ex then aluminium foil $\rightarrow$ small lengthscale also

Lumped capacitance model $\rightarrow$ can be used when Biot Number $\ll 1$

$$\frac{hL}{K} \rightarrow \begin{array}{l} h \rightarrow \text{small} \\ L \rightarrow \text{small} \\ K \rightarrow \text{large} \end{array}$$

Quenching of steel ball $\rightarrow$ centre will retain its temperature till last

First step of transient conduction $\rightarrow$ check Biot's number < 0.1



If yes then LC model is applicable

generally for liquids we will consider convection only, except highly conductive liquids like liquid metals (sodium) $\rightarrow$ coolant in nuclear reactor

9/3/23 Thursday

If work done on CV $\rightarrow$ energy increases

" " " against gravity $\rightarrow$ energy decreases
/ body force
or even surface forces like pressure

what if shear forces are also present $\rightarrow$ then work would also be need to done against it

Rate of work done = Force $\times$ Vol in direction of force

Viscous effects are important when velocity is very large and or system is very small

$\downarrow$
only then ϕ will appear in energy equation

$\downarrow$
i.e. velocity gradient is large

viscous dissipation $\rightarrow$ may not be relevant in most cases

Problem!

Lubricant in any bearing

liquid should not degrade $\rightarrow$ viscous breakdown

If h is very small as compared to R can be taken as flow between parallel plates $\rightarrow$ so linear velocity distribution

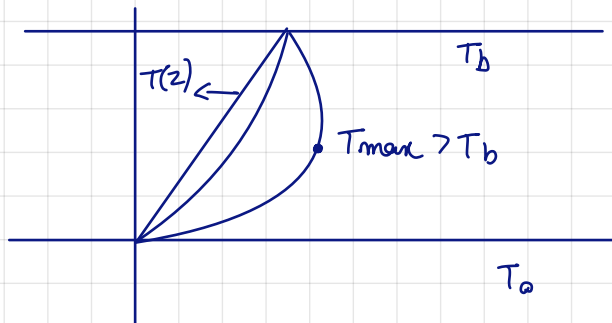
you have to solve momentum equation for energy equation

↳ One way coupling

Momentum equation $\rightarrow$ energy equation

Energy equation $\rightarrow$ Momentum equation (X)

$$Br = \frac{\text{viscous dissipation}}{\text{conductive heat transfer}}$$



For $Br > 2 \rightarrow$ there will be a maximum in temp profile which is greater than T_b

If $Br > 2 \rightarrow$ there will be a maximum temp in intermediate region

Problem 2

Transpiration cooling

Outward flow would reduce load on refrigeration coil

conduction $\rightarrow$ inward dirn] $\rightarrow$ opposite direction
convection $\rightarrow$ outside dirn

velocity will be a fn of $x \rightarrow$ as area keeps on changing

$$\underbrace{4\pi r^2}_{\text{area}} \times \underbrace{V_r}_{\text{velocity}} \times \underbrace{\rho}_{\text{density}} = \dot{W} \quad (\text{mass flow rate})$$

$R \rightarrow$ infinite $kR \rightarrow$ finite

↳ like a sphere kept in a room

you cannot have convection without conduction

effectiveness will depend on $\rightarrow$ flow rate geometry $R_o \rightarrow$ has thermophysical properties incorporated in it

we use transpiration when very high temp $\rightarrow$ rocket entering earth atmosphere
" low temp $\rightarrow$ cryogenics

ex $\rightarrow$ drying of shirt under fan

$\hookrightarrow$ convection through forced mass transfer through diffusion

Natural Convection

$\hookrightarrow$ buoyancy driven flow

$$\beta = \frac{1}{V} \frac{\Delta V}{\Delta T} \rightarrow \text{coefficient of volumetric expansion thermodynamic factor}$$

Buoyant force $\rightarrow$ per unit volume, as all terms in N-S eqn is per unit volume

Boussinesq $\rightarrow$ ρ remains the same, for convective part but for buoyant force we use $\bar{\rho}$] approximation

In forced convection $\rightarrow$ buoyant flow will be neglected
 $(-\bar{\rho} g \beta (T - \bar{T}))$

$$\rho(T) = \bar{\rho} - \bar{\rho} \beta (T - \bar{T})$$

$\bar{\rho} \rightarrow$ evaluated at $\bar{T} = (T_A + T_B)/2$

Nusselt no $\rightarrow$ non dimensional temperature gradient at interface

$h \rightarrow$ local heat transfer coefficient

h decreases with $x \rightarrow$ till laminar
once turbulent flow starts $\rightarrow$ more mixing $\rightarrow h$ shoots up

